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Vaccination with S_{pan} , an antigen guided by SARS-CoV-2 S protein evolution, protects against challenge with viral variants in mice

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SARS-CoV-2 continues to accumulate mutations to evade immunity, leading to breakthrough infections after vaccination. How researchers can anticipate the evolutionary trajectory of the virus in advance in the design of next-generation vaccines requires investigation. Here, we performed a comprehensive study of 11,650,487 SARS-CoV-2 sequences, which revealed that the SARS-CoV-2 spike (S) protein evolved not randomly but into directional paths of either high infectivity plus low immune resistance or low infectivity plus high immune resistance. The viral infectivity and immune resistance of variants are generally incompatible, except for limited variants such as Beta and Kappa. The Omicron variant has the highest immune resistance but showed high infectivity in only one of the tested cell lines. To provide cross-clade immunity against variants that undergo diverse evolutionary pathways, we designed a new pan-vaccine antigen (S_{pan}). S_{pan} was designed by analyzing the homology of 2675 SARS-CoV-2 S protein sequences from the NCBI database before the Delta variant emerged. The refined S_{pan} protein harbors high-frequency residues at given positions that reflect cross-clade generality in sequence evolution. Compared with a prototype wild-type (S_{wt}) vaccine, which, when administered to mice, induced serum with decreased neutralization activity against emerging variants, S_{pan} vaccination of mice elicited broad immunity to a wide range of variants, including those that emerged after our design. Moreover, vaccinating mice with a heterologous S_{pan} booster conferred complete protection against lethal infection with the Omicron variant. Our results highlight the importance and feasibility of a universal vaccine to fight against SARS-CoV-2 antigenic drift.

INTRODUCTION

The outbreak of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) in late 2019 has caused a global pandemic that is still ongoing (1–3). Worldwide, as of 30 July 2022, 559,469,605 confirmed cases of coronavirus disease 2019 (COVID-19), including 6,361,157 deaths, had been reported to the World Health Organization (WHO) (4). SARS-CoV-2 has a relatively high mutation frequency as an RNA virus with a large genome size. Various mutations have appeared after more than 2 years of continuous transmission (5), and mutations in the spike (S) protein have changed the infectivity and antigenicity of the virus (6). The first prevalent mutation was D614G, which altered the fusion-competent conformation of the S protein and enhanced angiotensin-converting enzyme 2 (ACE2) binding ability (7). Subsequently, the A222V mutation, first detected in Spain, spread quickly throughout Europe in the middle of 2020 (8). As different mutations occur constantly, the term “variant” has been used to denote viral strains that

carry a series of emerging amino acid mutations (9). The WHO subsequently announced a Greek alphabet–based naming scheme for variants, including variants of concern (VOCs) and variants of interest (VOIs) (10).

The Alpha variant (B.1.1.7) was documented in December 2020, and globally, more than 80% of new infections were identified as the Alpha variant between April and May 2021 (11). The Beta variant (B.1.351) was detected after the Alpha variant (B.1.1.7) and became the dominant variant in South Africa within 1 month (12). The Delta variant (B.1.617.2) was identified in October 2020 in India and spread faster than other variants (60% more transmissible than the Alpha variant) (13). However, to date, more than 96% of new infections worldwide have been identified as Omicron subvariants (14). Although the progression of variants seems largely unpredictable, there are apparently some underlying trends in virus evolution.

Virus mutagenesis appears to either enhance infectivity for wider spread, escape preexisting immunity for survival, or both (15). Viral glycoprotein mutations that enhance infectivity without modulating immune escape activity have minimal effects on vaccine effectiveness. For instance, the D614G mutation on the S protein of the SARS-CoV-2 Alpha variant is proven to enhance viral infectivity by about 2.5- to 9.3-fold (16), but vaccines against the ancestral virus could still neutralize the D614G virus as efficiently as the wild-type (WT) virus (17). Another example is the Delta variant, whose infectivity is vastly enhanced and was once the principal prevalent strain (18). Again, many studies demonstrated

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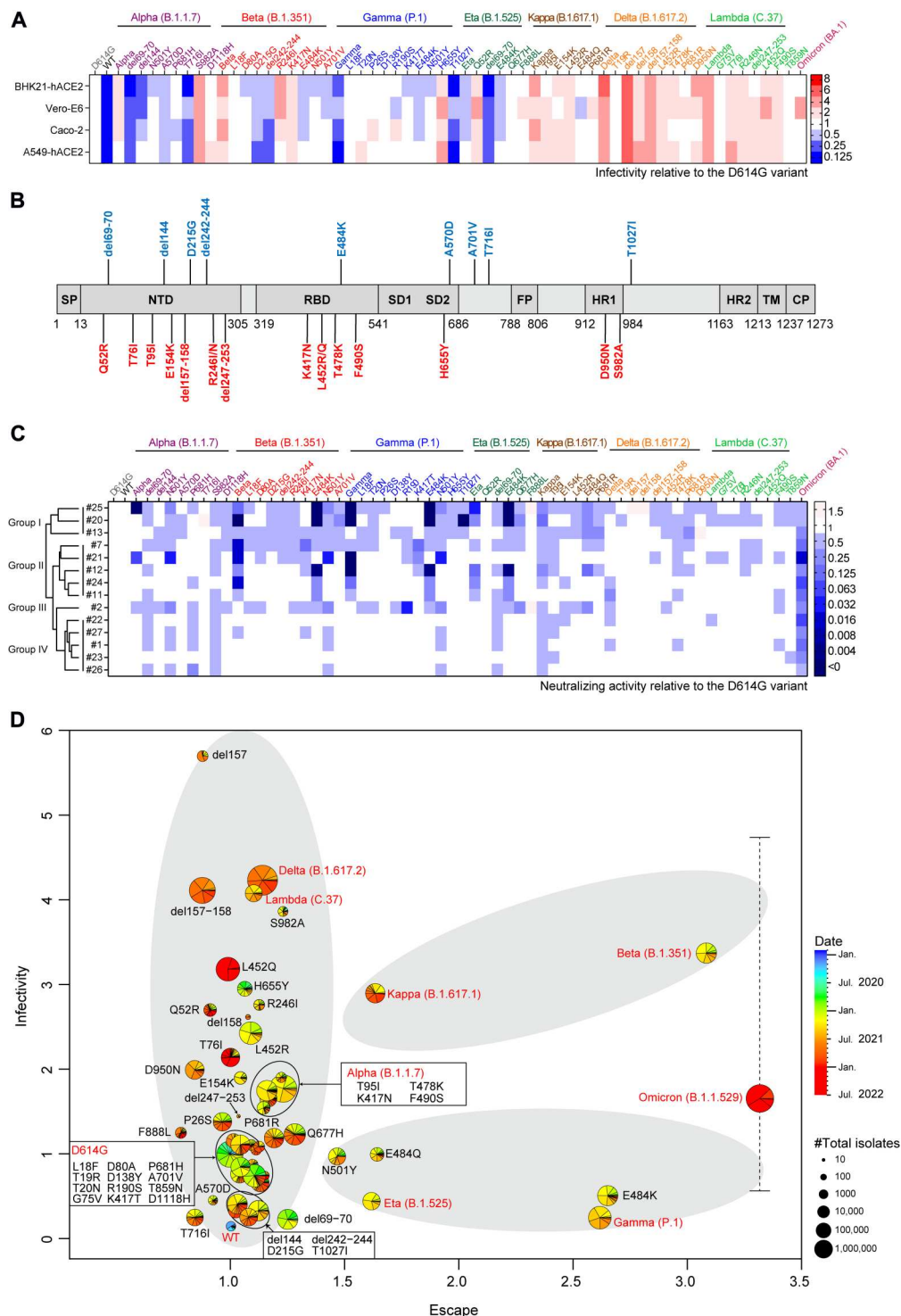
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that prototype vaccines maintained their effectiveness against the Delta variant (19–21). In contrast, some mutations have no effects on infectivity but increase viral immune escape, dampening the vaccine's effectiveness. For example, E484K and E484Q mutants evade the immunity of prototype vaccines by 2.2- to 7.1-fold and 1.8- to 4.5-fold, respectively (22, 23). The Gamma variant harboring E484K was highly immune resistant, decreasing the prototype vaccine's

neutralization activity by 4.5- to 6.7-fold (24). The most threatening mutation is to increase both infectivity and immune escape. The Beta [$R_0 = 4$ to 5 (25)] and Omicron [$R_0 = 9.5$ to 18.6 (26)] variants are more infectious and transmissible than the ancestral virus. In addition, the Moderna (mRNA-1273) and Pfizer/BioNTech (BNT162b2) prototype mRNA vaccines were less effective at neutralizing the Beta variant (approximately 27.7- to 34.5-fold

Fig. 1. The infectivity and monoclonal antibody neutralizing activity of the SARS-CoV-2 variants reveal SARS-CoV-2 S protein evolutionary dynamics. (A) Infection assay results are shown for 54 SARS-CoV-2-related mutant pseudoviruses in four indicated cell lines, all of which are known to be susceptible to SARS-CoV-2. The infectivity of the D614G variant was set to 1. Blue and red represent decreased and increased viral infectivity compared with the reference D614G variant in the indicated cell lines. Data represent the means of three independent experiments. (B) Location of mutations in the S protein affecting viral infectivity. Crucial mutations were identified in secondary structures of the S protein. The color blue indicates decreased infectivity, and the color red indicates increased infectivity. SP, signal peptide; NTD, N-terminal domain; RBD, receptor binding domain; SD, subdomain; FP, fusion peptide; HR, heptad repeat; TM, transmembrane; CP, cytoplasm domain. (C) Heatmap representation of neutralizing activity for 14 monoclonal antibodies against 54 SARS-CoV-2-related mutant pseudoviruses; the neutralizing activity of the D614G variant was set as 1. Blue represents decreased neutralizing activity. Data represent the means of three independent experiments. The groups indicate antibodies classified based on the Complex Heatmap package method (34). Numbers on the y axis indicate antibody serial numbers as shown in table S1. (D) Diagram of SARS-CoV-2 evolution. The x axis shows the immune escape ability, and the y axis shows the infectivity of SARS-CoV-2 variants (the error bar indicates the diversity of cell-specific infectivity for Omicron). The color of the pie charts marks the isolation time of the indicated variants recorded in the GISAID database. Each sector's area size indicates the numbers of isolates of individual variants until July 2022 (per month). The gray shadows indicate the evolutionary paths.



reduction) (24); this effect was even more pronounced with the Omicron variant (up to 40-fold reduction in neutralization) (27, 28). Thus, a systematic analysis of viral mutagenesis is necessary to guide vaccine design for covering both infectious and resistant sites.

Recently, studies showed that homologous or heterologous prototype booster vaccination (29) or immune regulator adjuvant could elevate the overall neutralization activity against variants (30). However, antigens designed to be more resistant to any current variant and also variants in the future are urgently needed. No matter how mutated SARS-CoV-2 is in the human population, it all comes from the same ancestral strain and the same evolutionary root with similarities in sequence. As a result, an ideal universal vaccine antigen should take consensus sites of majority variants to promote broad neutralization of those and other variants. Therefore, redesigning and updating SARS-CoV-2 vaccines to use a universal antigen would be a next-generation strategy to protect against future variants.

In the present study, we defined the infectivity and immune escape ability of 54 SARS-CoV-2 pseudoviruses (covering 45 single mutations, eight variants, and the WT strain) and then analyzed the evolutionary route of the SARS-CoV-2 S protein. We found that the SARS-CoV-2 S protein evolved into distinct evolutionary paths of either high infectivity plus low immune resistance or low infectivity plus high immune resistance, resulting in heterogeneous antigenicity. Thus, the current single-strain vaccine could provide only narrow-spectrum protection to one group of variants but not others. Through phylogenetic analysis of 2675 S proteins, we generated a universal S_{pan} vaccine, taking the most frequent mutations that could elicit broadly potent neutralization to all the VOCs and VOIs to date. This study sheds light on a universal SARS-CoV-2 vaccine to fight against continuous viral mutagenesis.

RESULTS

SARS-CoV-2 variants have distinct infectivity activity in vitro

The frequencies of SARS-CoV-2 variants have changed markedly from the beginning of the COVID-19 pandemic up until April 2022 (fig. S1). Mutations in the S protein increased one after another, leading to questions about whether such changes have a predictable regularity. To explore the S protein functions of SARS-CoV-2 variants and mutants, we generated 54 pseudotyped viruses expressing either prototype S_{wt} or mutated S proteins (covering 45 single-site mutations and eight defined variants in a D614G background) (fig. S2). The infectivity assays were performed with four SARS-CoV-2-susceptible cell lines: BHK21-hACE2, Vero E6, Caco-2, and A549-hACE2 (31, 32). We found that the Alpha, Beta, Kappa, Delta, and Lambda variants displayed, on average, 1.77-, 3.38-, 2.90-, 4.24-, and 4.05-fold increases in infectivity compared with the reference D614G variant in the four cell lines, respectively (Fig. 1A and fig. S3). The Delta variant showed the highest infectivity. In contrast, the Gamma and Eta variants exhibited 4.02- and 2.25-fold decreases in infectivity as compared with the D614G variant. The emerging Omicron variant displayed high infectivity (4.74-fold increase) in the Vero E6 cell line but an average 1.58-fold decrease in the other three cell lines. Because Vero E6 cells support an endocytosis pathway for virus infection (33), the

results indicate a probable preference for the endocytic route of the Omicron variant.

Referring to the individual single-site mutations, del157 (5.70-fold increase compared with D614G) and S982A (3.88-fold increase compared with D614G) played epistatic roles in increasing viral infectivity. In contrast, del69-70 (4.39-fold decrease compared with D614G) and T716I (4.00-fold decrease compared with D614G) contributed the most to reducing viral infectivity. In summary, single mutations in the N-terminal domain (NTD) region of VOCs or VOIs could either increase or decrease the infectivity. However, mutations in the receptor binding domain (RBD), except for E484K, mainly enhanced viral infectivity (Fig. 1B). Mutations in the S2 subunit, such as S982A and D950N in the heptad repeat 1 (HR1) region, strongly enhanced viral infectivity, indicating that the S2 subunit also plays an essential role in regulating viral infectivity (Fig. 1B).

SARS-CoV-2 variants exhibit decreased susceptibility to monoclonal antibody neutralization

To study the immune escape abilities of S protein mutants, we tested all 54 pseudoviruses against a set of 14 commercial monoclonal neutralizing antibodies targeting either the WT S1 or the WT RBD (table S1). As a control, we confirmed that 14 of 27 commercial monoclonal antibodies could effectively neutralize the D614G variant (fig. S4). Viral sensitivities to neutralization were calculated by neutralizing percentages relative to the reference D614G variant under the 50% neutralization dilution titer (NT_{50}) to D614G for each antibody. By defining immune escape as a twofold decrease in neutralizing activity of a monoclonal antibody compared with the reference D614G variant, we divided the 14 monoclonal antibodies into four subgroups based on the Complex Heatmap package method (Fig. 1C) (34). Briefly, group I antibodies showed decreased neutralizing activity against almost all the single-site mutations and variants. Compared with group I, the antibodies in group II showed much higher neutralizing activities against several mutations in the NTD and S2, such as L18F, del69-70, A701V, and T1027I (average neutralizing activities increased 1.28-, 1.22-, 1.61-, and 2.08-fold compared with group I antibodies, respectively). However, group II antibodies showed little neutralization activity against the E484K (#7, #11, #12, and #24) or N501Y (#21) mutation. In contrast, the group III antibody exhibited a better response to the E484K mutation. Antibodies in group IV showed broad neutralization of various variants and single-site mutations.

Examination of the single-site mutant pseudoviruses showed that the mutations with the strongest immune escape abilities were E484K (2.65-fold reduction compared with the reference D614G in neutralizing activities against group I to III antibodies) and N501Y (1.47-fold reduction compared with the reference D614G in neutralizing activities against group I to III antibodies) (Fig. 1C). E484Q in the Kappa variant showed a weaker immune escape ability than E484K. In addition, the del69-70, S982A, and P681H mutations showed moderate immune escape abilities.

When referring to the variants, the average sensitivity of the Alpha variant to monoclonal antibodies was similar to that of the D614G variant (Fig. 1C). The Beta variant showed a high ability to escape group I to III antibodies (3.08-fold reduction compared with the reference D614G variant in neutralizing activities). The Gamma and Eta variants displayed a broader escape spectrum against group

I to III antibodies because of the E484K mutation (2.62- and 1.62-fold reduction, respectively). Of concern, the Delta and Lambda variants showed only a mild ability to escape all four types of antibodies. The Omicron variant showed the highest resistance among all the variants (3.32-fold reduction compared with the reference D614G variant in neutralizing activities against group I to III antibodies) and even gained resistance to group IV antibodies (Fig. 1C, far right).

Characterizing the evolution of the SARS-CoV-2 S protein

To understand the SARS-CoV-2 S protein evolutionary trend, we generated a quadrant diagram (Fig. 1D) combined with isolate numbers to present the infectivity (y axis) and immune escape (x axis) of the SARS-CoV-2 S protein on the basis of the results in Fig. 1 (A and C). The color in each pie chart indicates the different isolation times of the individual variant or mutation. The area size of each pie chart indicates the numbers of isolates from January 2020 to July 2022, recorded in the Global Initiative on Sharing All Influenza Data (GISAID) database (the larger the size is, the larger the number of isolates) (Fig. 1D). The comprehensive results from 11,650,487 S sequences showed that one set of mutations, including del157, del157-158, S982A, H655Y, R246I, Q52R, and L452R, tends to increase viral infectivity. Another set of mutations, such as E484K, N501Y, and E484Q, tends to increase antibody escape. Few mutations had an effect on both the infectivity and the immune escape ability of SARS-CoV-2.

Only the Beta and Kappa variants distributed on the diagonal had high infectivity and high immune resistance (Fig. 1D). Other variants evolved to have either increased infectivity (vertical) or high immune escape ability (horizontal). For instance, the Delta and Lambda variants appear in the y axis direction with infectivity factors of 4.25 and 4.10, respectively, but they all showed a low ability to escape neutralizing antibodies. In contrast, the Eta and Gamma variants showed strong escape ability plus low infectivity, appearing in the x axis direction. The highly concerning Omicron variant, located on the far x axis, had the highest immune escape factor of 3.32. Nevertheless, its infectivity increased only in Vero E6 cells and not in the other three cell lines, indicating restricted infectivity. The data suggest that, in most cases, the viral infectivity and immune escape ability of the S protein are contradictory. The high infectivity of the S protein is associated with weak virus immune escape ability. In contrast, highly resistant variants are associated with low infectivity.

S_{pan} , an antigen based on a universal S protein, was designed to be variant resistant

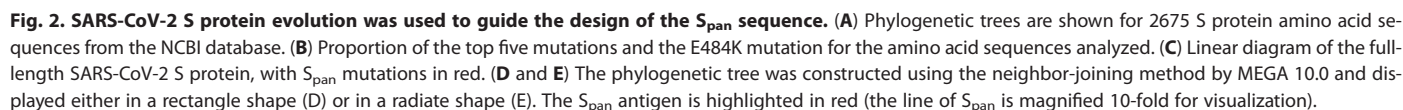
As shown in Fig. 1D, the S protein evolved in three distinct paths. Thus, the current single-strain vaccine strategy could protect against variants from the same evolutionary path but not the others. For instance, a WT S (S_{wt}) vaccine antigen cannot cover the Beta or Gamma variants as efficiently as the Alpha variant (35). To construct a pan-vaccine against cross-clade variants, we generated a universal S protein by downloading 2675 S protein amino acid sequences from the National Center for Biotechnology Information (NCBI) database (until 28 February 2021, before the Delta variant emerged) to build a phylogenetic tree using MEGA 10.0 software (Fig. 2A). The S protein sequences could be divided into five clades, of which clade 1 contains the most isolates of 2539 sequences, including variants from the B.1 descendants and Gamma

variants. Clades 2 and 3 contain 56 sequences, including a mixture of the Alpha, Beta, and Eta strains, among others. Clades 4 and 5 contain 80 sequences, including mutants other than the defined VOC or VOI nominates. Because the 2539 sequences in clade 1 are closely related and account for 95% of the sequences in the database, we calculated one consensus sequence for clade 1 and then added 136 sequences in clades 2 to 5 to obtain the frequency of concerning mutations. Five mutations—D614G, del69-70, del144, N501Y, and P681H—were the most common (Fig. 2B and table S2). In addition, considering the results of the neutralization experiment in Fig. 1C (where the E484K mutation conferred resistance to neutralizing antibodies), we designed a universal sequence of S_{pan} (E484K plus the above top five frequency mutations) (Fig. 2C) and another S_{15} (introducing nine more top mutations based on S_{pan}) (fig. S5). We further reconstructed phylogenetic trees using the eight VOCs (two isolate sequences each) and S_{pan} in either a rectangle shape (Fig. 2D) or a clock shape (Fig. 2E). The results showed that the S_{pan} sequence is located in the center of the phylogenetic tree, indicating its consensus characteristic. Although the Omicron variant emerged after we constructed S_{pan} , it clustered together with the S_{pan} sequence in the same clade, despite a far evolutionary distance from all of the other variants (Fig. 2, D and E). The data support that the S_{pan} sequence has cross-clade generality.

The S_{pan} vaccine antigen induces antibodies that can broadly neutralize SARS-CoV-2 variants

To test whether S_{pan} could serve as a pan-vaccine antigen against heterogeneous variants, we generated protein subunit vaccines by expressing the S trimer of either S_{wt} or S_{pan} in the Chinese hamster ovary (CHO-K1) eukaryotic expression system. Subsequently, both S_{wt} and S_{pan} antigens were mixed with adjuvants to formulate subunit vaccines, and then keratin 18 (K18)-hACE2 mice were immunized. Each mouse was inoculated with 25 μg per dose of antigen with primary and boost vaccination strategies at an interval of 14 days (Fig. 3A). Mouse serum samples were collected 14 days after the second vaccination.

Sera from the prototype S_{wt} -vaccinated mice had decreased neutralizing activities against all the VOCs or VOIs (Fig. 3B), especially the Omicron subvariants (43.4-fold average NT_{50} reduction to the reference D614G) and Beta variant (16.8-fold average NT_{50} reduction to the reference D614G). However, sera from S_{pan} -vaccinated mice exhibited a broader spectrum of neutralizing activities against all 10 variants and showed even more substantial neutralization of 6 of 10 variants (red color) (Fig. 3B and fig. S6). Furthermore, detailed serum neutralizing activities against all the individual mutations in VOCs and VOIs showed that sera from S_{pan} -vaccinated mice had better neutralization activity against immune-escaping mutations such as E484K, N501Y, E484Q, del69-70, F490S, P681H, and T478K (Fig. 3C). To our surprise, sera from S_{15} -vaccinated mice did not show better neutralizing activity against different variants than sera from S_{pan} -vaccinated mice (fig. S7), which implies that the design of vaccine antigens is not a simple superposition of mutation sites but requires joint analysis of rigorous bioinformatics calculations and experimental evidence.



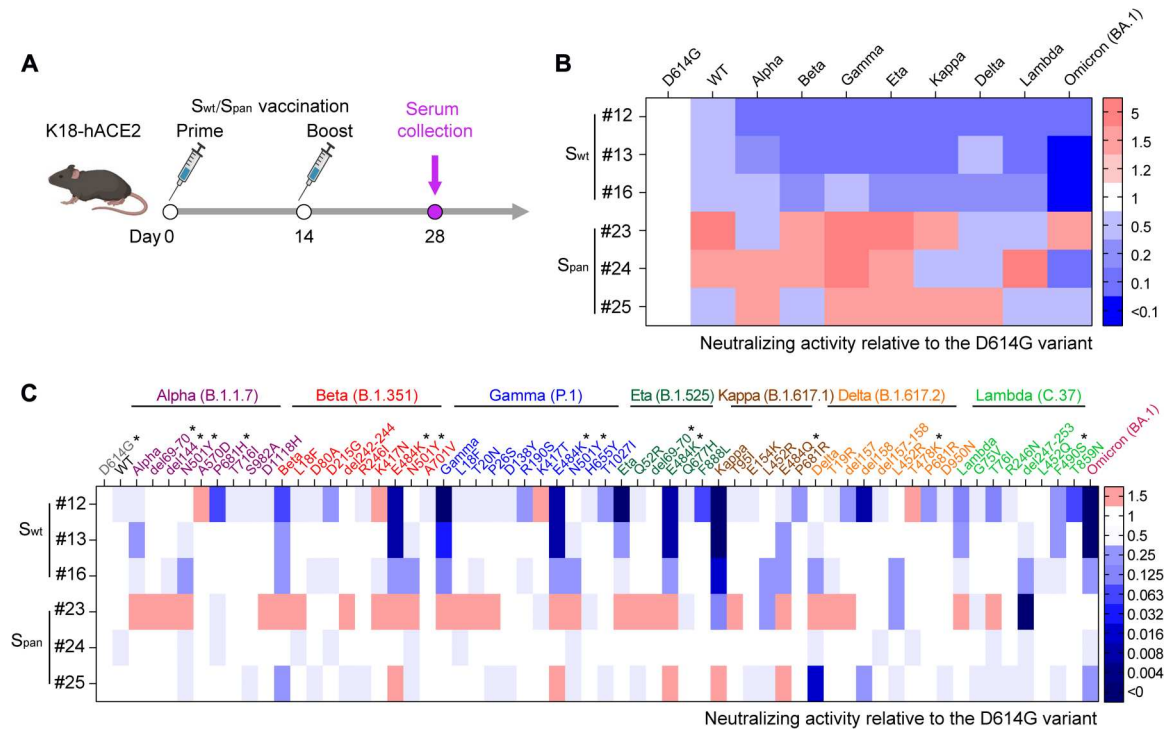


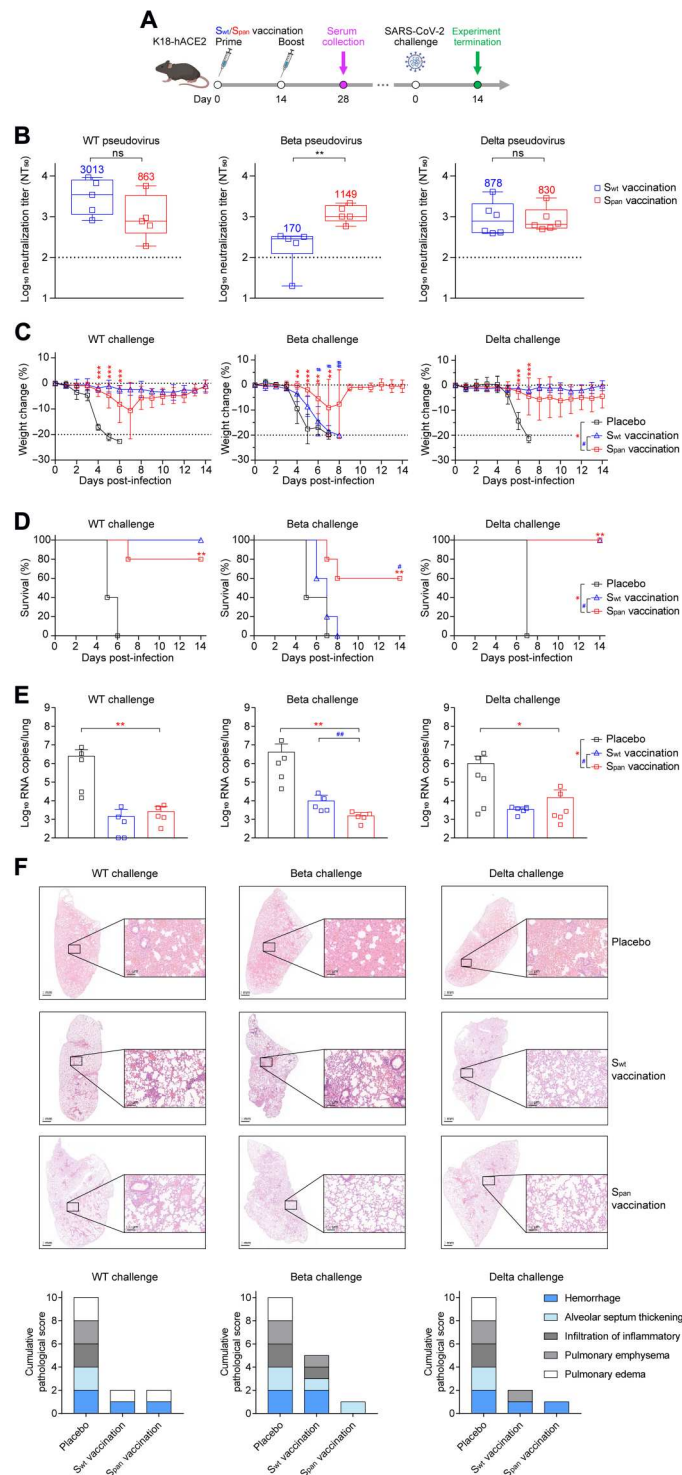
Fig. 3. Serum from *S_{pan}*-vaccinated mice neutralizes SARS-CoV-2 variant pseudoviruses. (A) Schematic diagram of the vaccination timeline. K18-hACE2 mice from each group were prime- and boost-vaccinated with either prototype *S_{wt}* or *S_{pan}* subunit vaccines (25 µg per dose) at days 0 and 14. Serum samples were collected on day 28. (B) Heatmap representation of the relative neutralizing activity of serum samples isolated from *S_{wt}*- and *S_{pan}*-vaccinated mice (*n* = 3 per group) against VOC and VOI pseudoviruses; neutralizing activity was determined by NT₅₀ values relative to the D614G variant. (C) Heatmap representation of neutralizing activity of one representative mouse serum sample per group against all 54 pseudoviruses. Relative neutralization was determined using a uniform antibody dilution fold of 1:1000. For (B) and (C), the neutralizing activity against the D614G variant was set to 1. Blue and pink represent decreased and increased neutralizing activity of serum samples, respectively. Data represent the means of three independent experiments. The asterisk (*) indicates the mutations that showed more resistance to neutralization by *S_{wt}* serum compared with *S_{pan}* serum. Data represent the means of three independent experiments.

***S_{pan}* vaccination protects K18-hACE2 mice from WT, Beta, and Delta SARS-CoV-2 challenge**

To evaluate *S_{pan}* vaccine protection against authentic viruses, we immunized K18-hACE2 mice (36) with a human-use dose of *S_{pan}* and *S_{wt}* vaccines (25 µg per dose) to achieve an adequate neutralization titer for this lethal challenge model. Phosphate-buffered saline (PBS) was used as a placebo control. After a primary and boost vaccination strategy, mice were grouped randomly and challenged with lethal doses of the prototype virus, the Beta virus, or the Delta virus by intranasal infection (Fig. 4A). Serum titers of neutralizing antibodies against the pseudotyped WT, Beta, and Delta viruses remained high, at geometric mean titer (GMT) values of 863, 1149, and 830, respectively, when mice were vaccinated with *S_{pan}* (Fig. 4B and fig. S8). In contrast, although high GMTs of 3013 and 878 for *S_{wt}* vaccination against the WT and Delta viruses were observed, titers against the Beta virus markedly decreased to 170. Upon challenge, the placebo group mice died from day 5 to day 7 after infection. At the same time, *S_{wt}* vaccination entirely lost protection upon the Beta virus challenge, although it could protect mice against the WT and Delta viruses (Fig. 4, C and D). In contrast, the *S_{pan}* vaccination conferred protection against the WT and Delta viruses (80 and 100% protection, respectively) and significantly (*P* < 0.05) better protection (60%) against the Beta virus than that of *S_{wt}* vaccination (0%). The lungs of mice were isolated from each group when individual death occurred or at

14 days after infection. Consistent with the body weight and survival data, although both *S_{wt}* and *S_{pan}* vaccination could reduce viral RNA load in WT- and Delta-infected lung tissues, *S_{pan}*-vaccinated mice showed lower viral RNA load than *S_{wt}*-vaccinated mice upon the Beta virus challenge in their lung tissues (Fig. 4E). For histopathological evaluations, infection with the three SARS-CoV-2 variants induced severe lung lesions in the lungs of placebo-vaccinated mice, characterized by denuding epithelial cells in large airways and thickening of alveolar septa surrounded by a mixed inflammatory cell population. The lungs of *S_{wt}*-vaccinated mice were also injured upon Beta virus challenge. In contrast, *S_{pan}*-vaccinated mouse lungs showed fewer pathological changes, resulting in reductions in cumulative pathology scores (Fig. 4F). Our results confirmed that *S_{pan}* vaccination conferred broad protection against SARS-CoV-2 variants that evolved in distinct paths. Moreover, *S_{pan}* can even elicit effective neutralizing antibodies against the Omicron subvariants at the GMT of 180 to 2333 that emerged after our design (fig. S9). These data suggest that the *S_{pan}* immunogen could serve as a pan-vaccine to protect against potential antigen drift of SARS-CoV-2 in the future.

Fig. 4. The S_{pan} subunit vaccine protects against infection with three SARS-CoV-2 variants in mice. (A) Schematic diagram of the vaccination and virus challenge timeline. K18-hACE2 mice from each group were prime- and boost-vaccinated S_{WT} or S_{pan} subunit vaccines (25 μ g per dose) or PBS (placebo) on days 0 and 14. Serum samples were collected on day 28. After vaccination, mice were challenged with three different variants of SARS-CoV-2 by intranasal infection. Body weight and survival were monitored for 14 days after infection or until a more than 20% reduction in body weight (WT challenge, $n = 5$; Beta challenge, $n = 5$; Delta challenge, $n = 6$). **(B)** NT_{50} values for individual S_{pan} - or S_{WT} -vaccinated mouse serum samples against the prototype, Beta, and Delta pseudoviruses. The numbers above the box plots indicate the geometric mean titers (GMTs). The box-and-whisker plots show the minimum, first quartile, median, third quartile, and maximum values. The dotted horizontal line indicates the lower limit of detection. An arbitrary value of 20 was assigned to the samples with values below the limit of detection for which it was not possible to interpolate the titer. **(C and D)** Body weights (C) and survival curves (D) of the mice after challenge with the indicated SARS-CoV-2 variant. The dotted horizontal line in (C) indicates a 20% reduction in weight. Body weights are presented as the mean percentage of weight change $\pm$ SD. **(E)** The viral genome copies of the SARS-CoV-2 *N* gene were quantified. Values represent the means $\pm$ SD of five or six individual mice. **(F)** Histopathologic studies were performed on lung samples from the indicated groups, and pathological scores were calculated. Scale bars, 2 mm or 100 μ m. Statistical analysis was conducted using Mann-Whitney tests for bar graphs (B and E), two-way ANOVAs for weight curves (C), and log-rank tests for survival curves (D). Asterisks (*) indicate significance between S_{pan} and S_{WT} vaccination in (B) or between placebo and S_{pan} vaccination in (C) to (E). Pound signs (#) indicate significance between S_{pan} and S_{WT} vaccination in (C) to (E). * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$. ns, not significant.



S_{pan} heterologous booster vaccination enhances the ability to neutralize Omicron subvariants and protects K18-hACE2 mice from BA.1 challenge

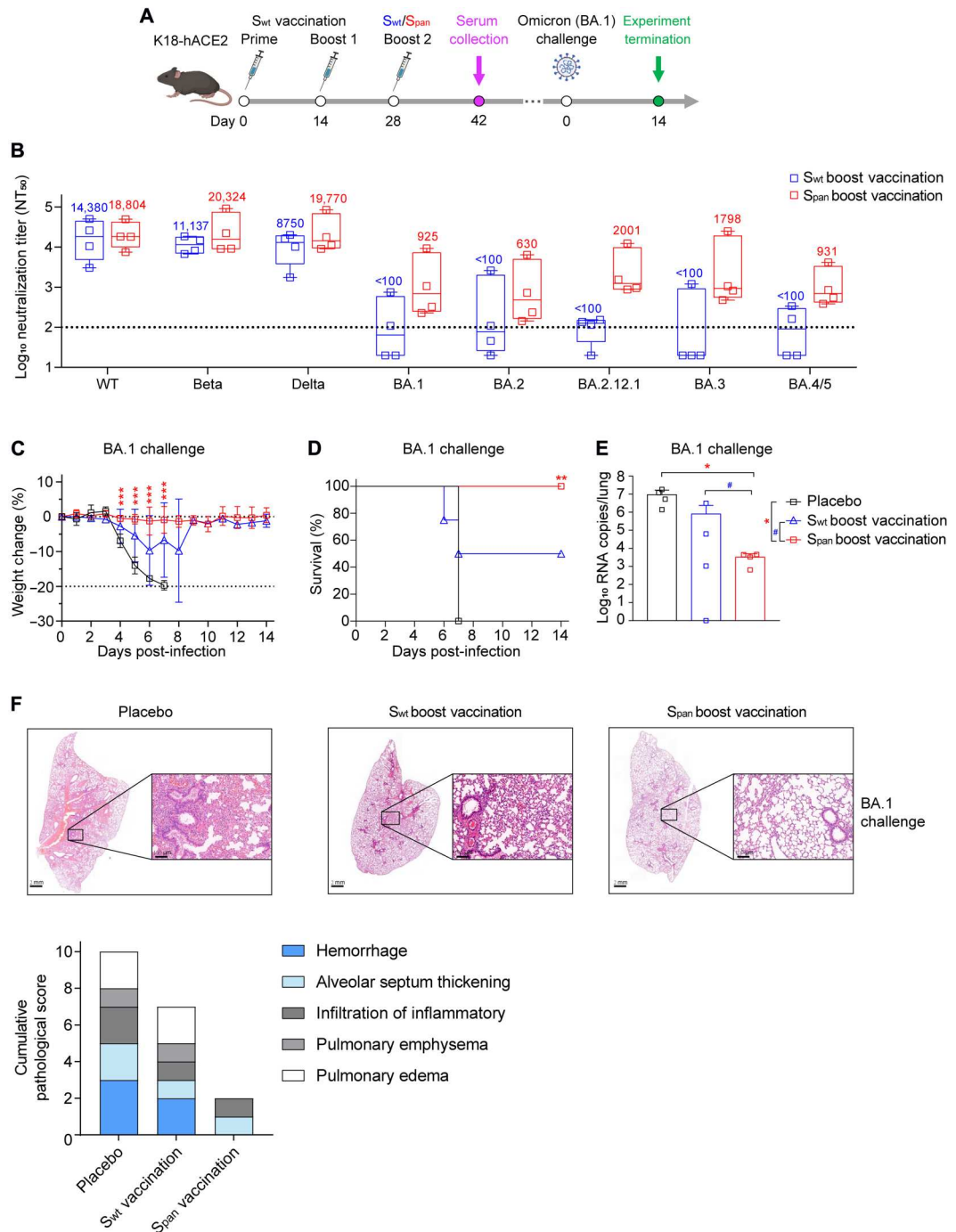
Because most primary vaccinations to date are based on the WT S protein, we evaluated the immunogenicity and protection of the S_{pan} as a heterologous booster. K18-hACE2 mice were booster-vaccinated with S_{pan} or S_{WT} after two doses of the S_{WT} vaccine. PBS was

used as a placebo control. After completing three vaccination doses, K18-hACE2 mice were intranasally infected with a lethal dose of authentic BA.1 virus (Fig. 5A).

Our results showed that three-dose booster vaccination with either S_{pan} or S_{WT} could elicit higher neutralization titers than two doses against different variants (WT, Beta, and Delta pseudoviruses; Fig. 5B and fig. S10). However, only S_{pan} booster vaccination could

Fig. 5. S_{pan} boost vaccination protects mice from BA.1 variant challenge. (A) Schematic diagram of the vaccination and virus challenge time-line.

K18-hACE2 mice were boost-vaccinated with S_{wt} or S_{pan} subunit vaccines (25 μg per dose) after two doses of the S_{wt} vaccine. Three doses of PBS were used as placebo controls. Serum samples were collected on day 42. After vaccination, mice were challenged with the BA.1 variant of SARS-CoV-2 by intranasal infection. Body weight and survival were monitored for 14 days after infection or until a more than 20% reduction in body weight ($n = 4$ in each group). (B) NT_{50} values of individual serum samples isolated from S_{pan} or S_{wt} boost-vaccinated mice sera against the indicated SARS-CoV-2 variants ($n = 4$). The numbers above the box plots indicate the GMTs. The box-and-whisker plots show the minimum, first quartile, median, third quartile, and maximum values. The horizontal dotted line indicates the lower limit of detection. An arbitrary value of 20 was assigned to the samples with values below the limit of detection for which it was not possible to interpolate the titer. (C and D) Body weights (C) and survival curves (D) for the indicated groups of mice after BA.1 challenge. The dotted horizontal line in (C) indicates a 20% reduction in weight. Body weights are presented as the mean percentage of weight change $\pm$ SD. (E) The viral genome copies of the SARS-CoV-2 *N* gene were quantified. Values represent the means $\pm$ SD of four individual mice. (F) Histopathologic studies were performed on lung samples from the indicated groups, and the pathological score was quantified. Scale bars, 2 mm or 100 μm . Statistical analysis was conducted using a two-way ANOVA for weight curves (C), a Mann-Whitney test for bar graphs (E), and a log-rank test for survival curves (D). Asterisks (*) indicate significance between placebo and S_{pan} boost vaccination, and pound signs (#) indicate significance between S_{pan} and S_{wt} boost vaccination. * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.



elicit effective neutralization antibodies against Omicron and its subvariants; GMTs from S_{pan} booster were 925, 630, 2001, 1798, and 931 against the BA.1, BA.2, BA.2.12.1, BA.3, and BA.4/5 variants, respectively (Fig. 5B). In contrast, the GMTs were all less than 100 in the S_{wt} homologous booster group (Fig. 5B). Upon BA.1 lethal challenge, the mice from the placebo group exhibited a precipitous body weight loss and 100% mortality, and the S_{wt} homologous boost vaccination could provide only 50% protection.

Conversely, S_{pan} booster vaccination could provide 100% protection against BA.1 (Fig. 5, C and D). Consistent with the body weight and survival data, S_{pan} booster vaccination reduced viral RNA loads 2846-fold in BA.1-infected mouse lungs compared with placebo vaccination (Fig. 5E). In contrast, viral RNA loads in the S_{wt} booster groups were still detectable at high concentrations in animals who succumbed to infection. Histopathologically, the lungs of mice from the placebo group and S_{wt} booster group

exhibited alveolar damage, inflammatory cellular infiltration, and hemorrhage. In contrast, mice in the S_{pan} booster group had no apparent lesions (Fig. 5F). These results demonstrate that the S_{pan} vaccine could be used as a sequential booster to elicit a broad neutralization titer against VOCs, including the evolutionarily distant Omicron subvariants.

DISCUSSION

The emergence of various SARS-CoV-2 variants has become a major concern during the pandemic. More than 7300 types of amino acid substitutions, deletions, and insertions have been found in the SARS-CoV-2 S protein. Mutations are categorized into two directions. One type makes the virus more infectious with high cell binding and entry ability, leading to increased transmissibility in the human population. The other type enables the virus to evade the host's immune responses, such as neutralizing antibodies, leading to breakthrough infections after vaccination (31, 37, 38). In particular, L452R has been reported to increase the infectivity of the Delta variant (39). For immune escape ability, E484 is the most critical amino acid. Changes in the E at position 484 to K, Q, or A reduced the neutralizing activity of antibodies by more than 10-fold (40, 41). How SARS-CoV-2 will evolve in the future is still unknown, which largely influences the effectiveness of current vaccines.

In this study, the infectivity of 54 pseudotyped SARS-CoV-2 viruses with individual mutations was systemically studied. We found that the most effective mutations were mainly enriched in the NTD and RBD, whereas mutations outside these two domains had minimal effects on viral infectivity (Fig. 1A). To our surprise, the S982A mutation, which is located in the HR1 domain of the S2 subunit, showed a high ability to increase viral infectivity. The HR1 domain is highly conserved among human coronaviruses and plays an essential role during the viral membrane fusion process, serving as a broad-spectrum drug target (42, 43). In addition, S982A has decreased sensitivity to neutralization by group IV antibodies, which are broadly reactive to other mutations, indicating that the S982A mutation should be considered if it frequently reemerges.

The results from our immune escape analysis are similar to those of published studies in that the Beta and Gamma variants showed strong resistance to neutralizing antibodies. Both the Beta and Gamma variants had E484K and N501Y mutations in the RBD that contributed to evading immunity. However, E484K and N501Y could not increase viral infectivity compared with the reference D614G mutant, suggesting that a particular mutation's infectivity and immune escape ability may be incompatible. Nevertheless, combinations of several mutations in the Beta variant increased both the infectivity (contributed by R246I and K417N) and the immune resistance ability (contributed by E484K and N501Y) to make more substantial gains. However, such variants gaining dual functions, such as the Beta variants, are limited to a relatively small number of isolates. In contrast, the Delta variant, with a large number of isolates, showed the highest infectivity but only a mild immune escape ability, indicating that the virus may have to sacrifice immune escape potential to obtain higher infectivity. Unexpectedly, the Omicron variant emerged during the preparation of our manuscript. The analysis of Omicron showed the strongest immune escape ability, even stronger than that of the Beta variant, probably contributed by its many

mutations. However, the infectivity of Omicron was different from that of the Beta variant, which exhibited an average increase in infectivity in all four tested cell lines. The Omicron variant was more infectious in Vero E6 cells but less infectious in BHK21-hACE2, Caco-2, and A549-hACE2 cells. We suspect that the emerging variants of Omicron might struggle to balance infectivity and immune evasion, so the evolution of the Omicron variant is still ongoing.

SARS-CoV-2 mainly depends on two pathways, the transmembrane protease serine 2 (TMPRSS2)-mediated membrane fusion pathway and the cathepsin B/L-mediated endocytic pathway, to enter cells (44). Previous studies showed that SARS-CoV-2 entry into Vero E6 cells occurred mainly through the endocytic pathway due to the low expression of TMPRSS2 in these cells (33). This finding might indicate that the Omicron variant prefers the endocytic pathway to enter cells. It has been shown that TMPRSS2 is highly expressed in alveolar type I and II cells of the lung (45), so the Omicron variant might be relatively less infectious in these cells. More studies are needed on susceptibility of different cell types because of changes in the viral entry pathway.

Because the SARS-CoV-2 S protein evolved in three distinct directions (Fig. 1D), a universal vaccine to provide broad protection must consider all the heterogeneous mutations within the same antigen. Such a generality does not imply a simple averaging of mutations, but reasonable calculations must be done based on the evolutionary trajectory of the antigen. Here, we generated a candidate universal S protein antigen (S_{pan}) by phylogenetically calculating a collective, universal sequence including the most frequent mutations from five different clades. The inclusion of high-frequency mutations in the S_{pan} vaccine led to protection against highly infectious variants, such as the Delta and Lambda variants, after vaccination. In contrast, high immune escape variants were covered by mutations, including E484K. Our results showed that the S_{pan} antigen could elicit broad humoral immunity against heterogeneous variants, including the WT, Beta, and Delta variants.

Our S_{pan} antigen not only improved the protection against the covered mutations but also elicited better neutralizing activity against uncovered mutations, such as E484Q, T478K, and F490S (Fig. 3C). Consistent with this, group III and IV antibodies also showed broad-spectrum neutralizing activity against these mutations, suggesting that conserved epitopes might exist in the SARS-CoV-2 S protein. We further tested the neutralization of the S_{pan} vaccine against the Omicron variant that emerged after our initial antigen design. Our results showed that the Omicron variant showed mild resistance to S_{pan} -vaccinated mouse serum (sixfold reduction). In comparison, the neutralizing activity of sera from mice vaccinated with the prototype strain vaccine was reduced approximately 22- to 41-fold against the Omicron variant (28, 46). Our S_{pan} serum showed a better ability to neutralize not only the variants circulating at the time of its generation but also a variant that arose afterward, once again supporting the generalizability of the S_{pan} immunogen.

Recent studies revealed that the heterologous boost strategy could enhance the background neutralization ability induced by the prototype strain vaccine (47–49). However, this strategy was not fully effective against Omicron subvariant infection (50). Our study showed that boosting with S_{pan} led to higher Omicron-specific neutralization antibodies than boosting with S_{wt} . This finding might be due to the consensus characteristics of S_{pan} . Thus, after two doses of prototype strain vaccines, a third dose of the broad-reactive S_{pan}

vaccine could serve as a superior heterologous booster against Omicron subvariant infection.

To date, multiple broad-spectrum antigens based on various platforms are under development against SARS-CoV-2. However, most antigens are based on chimeric sequences from circulating variants that cannot cover every variant and cannot predict future variants (51, 52). In our strategy, the universal antigen S_{pan} was obtained by calculating consensus sequences based on the evolutionary history of the SARS-CoV-2 S protein. This unique strategy allows S_{pan} to achieve a broad-spectrum activity to neutralize various circulating strains, as well as resistance to future variants.

Our study has several limitations. First, the *in vivo* study was performed only in mice and was limited by a relatively small number of animals per group. It will be essential to include additional animal models before translating this vaccine to clinical trials. Second, the durability of immune responses needs to be further determined, especially for the heterologous booster vaccination. Third, the S_{pan} antigen was designed before the Delta variant emerged, although it was capable of eliciting an immune response against the Delta variant. Similarly, although the current Omicron variant differs in antigenicity from the Delta variant both in our and other studies (53), the S_{pan} vaccination could still neutralize the Omicron subvariants *in vitro*. Because Omicron BA.1 has been replaced by the BA.5 and other new subvariants, future experiments challenging animals with these more recent variants are important to determine how broad of an immune response S_{pan} may elicit.

In conclusion, our pan-vaccine antigen strategy conferred protection against cross-clade variants, including vaccine-resistant Omicron variants, which highlights a potential opportunity to combat SARS-CoV-2 antigenic drift. Thus, redesigning the S protein antigen and optimizing our vaccination strategies could confer a substantial protection advantage, which might finally lead to broad protection against existing and future SARS-CoV-2 variants.

MATERIALS AND METHODS

Study design

The primary aim of this study was to study the broad immune response and protective effects of the S_{pan} vaccine against SARS-CoV-2 variants. The sample sizes were empirically estimated by considering the variations of the results and the statistical power needed while minimizing the number of animals. Female BALB/c mice were randomly assigned to groups of five mice to determine the antibody response induced by S_{pan} . Mixed-sex K18-hACE2 mice were randomly assigned to groups of three mice to determine the humoral immune response and were randomly assigned to groups of four to six mice to determine S_{pan} protection against SARS-CoV-2 variants. Studies were not blinded. No data points were omitted from the analysis. Animal studies were approved by the Institutional Animal Care and Use Committee (WP20210380 and WP20220098).

Cell lines

The A549-hACE2, Caco-2, and Vero E6 cell lines were obtained from the American Type Culture Collection. Cells were maintained in Dulbecco's modified Eagle's medium (DMEM; Gibco) supplemented with 10% fetal bovine serum (FBS). H. Yan (Wuhan

University) provided the BHK21-hACE2 cell line. All cells were incubated at 37°C in 5% CO₂.

Viruses and murine infections

The WT, Beta, Delta, and BA.1 SARS-CoV-2 viruses were supplied by Hubei Center for Disease Control and Prevention and multiplied in Vero E6 cells to make virus stock. For mouse infection, 250 plaque-forming units of the WT, Beta, and Delta viruses and 1000 plaque-forming units of the BA.1 virus were used to intranasally infect K18-ACE2 mice in 50 µl of PBS. All infections were conducted in the A3 laboratory of Wuhan University. The mice were then monitored daily for changes in their body weight and survival. The Animal Committee of Wuhan University has approved the corresponding animal experiments.

Plasmid construction and site-directed mutagenesis

The DNA sequences of human codon-optimized S proteins from SARS-CoV-2 variants (B.1.1.7, GISAID: EPI_ISL_601443; B.1.351, GISAID: EPI_ISL_678597; P.1, GISAID: EPI_ISL_906075; B.1.525, GISAID: EPI_ISL_1093472; B.1.2, GISAID: EPI_ISL_2000505; B.1.617.1, GISAID: EPI_ISL_1660428 S1; B.1.627.2, GISAID: EPI_ISL_2029113; C.37, GISAID: EPI_ISL_3023383; B.1.1.529, GISAID: EPI_ISL_7162071) and S protein mutations were commercially synthesized or generated by overlapping polymerase chain reaction (PCR)-based mutagenesis. pCAGGS-SARS-CoV-2-S-C9 (gifted from W. Li, National Institute of Biological Science, Beijing, China) was used as a template and cloned into the pCAGGS vector. C-terminal 18-amino acid truncation was used to improve vesicular stomatitis virus (VSV) pseudotyping efficiency (54).

Pseudovirus production

Pseudotyped VSV-ΔG viruses expressing a luciferase reporter were provided by N. Xia (Xiamen University). To produce pseudotyped VSV-ΔG-Luc bearing SARS-CoV-2 S protein (pseudo-SARS-CoV-2), we seeded Vero E6 cells in 10-cm dishes and transfected them simultaneously with 15 µg of SARS-CoV-2-S-Δ18 plasmid using Lipofectamine 2000 (Thermo Fisher Scientific). Forty-eight hours after transfection, 500 µl of pseudotyped VSV-ΔG bearing VSV-G protein was added to the transfected Vero E6 cells. Cell supernatants containing pseudovirus were collected after another 24 hours. Cell debris was cleared by centrifugation at 3000 rpm for 6 min, and then samples were aliquoted and stored at −80°C.

Titration of the pseudoviruses

The day before pseudovirus titration, BHK21-hACE2 cells were seeded into 96-well culture plates at an appropriate density. To titrate the pseudoviruses, we made an initial 10-fold dilution in infection medium (DMEM, supplemented with 2% FBS and 1% penicillin/streptomycin) with six biological replicates in 96-well plates, followed by serial threefold dilutions for a total of nine dilutions. The last column of the 96-well plate served as the cell control without pseudoviruses. Then, the diluted pseudoviruses were transferred to the target cells and incubated at 37°C in a humidified atmosphere with 5% CO₂. After incubation for 24 hours, the supernatant was removed, and the cells were washed with cold PBS, followed by lysing in 40 µl of cell lysis buffer (Promega, E1941). Then, 20 µl of the lysate and 10 µl of luciferase substrate (Promega, E4550) were added into white 96-well plates for

chemiluminescence detection using the Varioskan LUX Multimode Microplate Reader (Thermo Fisher Scientific, VLBLATGD2). The 50% tissue culture infectious dose (TCID₅₀) of the pseudoviruses was calculated according to the Reed-Muench method. In addition, wells with a relative luminescence unit (RLU) value equal to or higher than 10,000 were considered positive (55).

Pseudovirus infection assay

The day before pseudovirus infection, the targeted cells were seeded into 96-well culture plates at an appropriate density. Using the lowest TCID₅₀ as a reference, we diluted the pseudoviruses to the same TCID₅₀ with the infection medium. After normalization, 100 µl of the pseudovirus with twofold dilution was added into the targeted wells in 96-well cell culture plates. After a 24-hour incubation at 37°C with 5% CO₂, the cells were washed with cold PBS and lysed in 40 µl of cell lysis buffer (Promega, E1941). Chemiluminescence detection was performed to detect the RLU values as described in the titration of pseudoviruses. Each group contained three biological replicates, and these experiments were performed three times independently.

Neutralization assays

The day before neutralization assays, the BHK21-hACE2 cells were seeded into 96-well culture plates at an appropriate density. The SARS-CoV-2 pseudoviruses were incubated with serially diluted serum samples at 37°C for 1 hour in 96-well white flat-bottom culture plates and then mixed with BHK21-hACE2 cells. After 24 hours, the cells were lysed in 40 µl of cell lysis buffer (Promega, E1941). Then, 20 µl of the lysate and 10 µl of luciferase substrate (Promega, E4550) were added into white 96-well plates for chemiluminescence detection using a Varioskan LUX multimode microplate reader (Thermo Fisher Scientific, VLBLATGD2). NT₅₀ was calculated by GraphPad Prism 7 (GraphPad Software Inc.) software with nonlinear regression curve fitting (normalized response, variable slope). GMT values were calculated according to NT₅₀.

SARS-CoV-2 S protein expression and purification

The DNA sequence encoding SARS-CoV-2 S residues 1 to 1208 (GenBank MN908947 nucleotides 21563 to 25384) was expressed in the CHO-K1 cells by codon optimization. We constructed the sequences carrying 6 mutations (S_{pan}) and 15 mutations (S₁₅) through rigorous bioinformatics calculation. The sequences were synthesized and constructed between the Hind III and Not I site into the GS-KS001 plasmid (Quacell Biotechnology). In addition, an 8×His tag was further added to the protein C terminus to facilitate protein purification. Then, the CHO-K1 cells were transfected with the S_{wt}, S_{pan}, or S₁₅ plasmids by Bio-Rad Gene Pulser Xcell system. The three proteins were fused with His tag and purified by Ni-NTA columns (GenScript).

Enzyme-linked immunosorbent assay of SARS-CoV-2 S protein

The purified S_{wt}, S_{pan}, or S₁₅ proteins were quantified by enzyme-linked immunosorbent assay. Anti-S antibodies (Sino Biological, 40150-D006) were coated on 96-well plates (Costar, 3590) overnight at 4°C with 100 µl of streptavidin (50 µg/ml) diluted in 0.1 M carbonate buffer (pH 9.6). Plates were subsequently washed three times with 0.05% PBST (0.05% Tween 20 in PBS). Excess binding sites were blocked using 200 µl of 0.05% PBST containing 5% bovine

serum albumin (BSA) for 2 hours at 37°C. Afterward, commercial S proteins (SinoBiological, 40589-V08B1) were used as standard sample and diluted in concentrations from 20,000 to 4.88 ng/ml at two times dilution ratio in 0.05% PBST containing 1% BSA. Meanwhile, S_{wt}, S_{pan}, or S₁₅ proteins were prepared in 0.05% PBST containing 1% BSA at dilutions of 1:100 to 1:10000. Subsequently, the diluted proteins were added to the coated 96-well plates (100 µl per well) at 37°C for 2 hours with three parallel wells in each group. Then, the plates were washed three times with 0.05% PBST, followed by the addition of the anti-S horseradish peroxidase-conjugated antibody (Sino Biological, 40150-D001-H) for 1 hour. The plates were again washed three times with 0.05% PBST, and the remaining peroxidase activity in the wells was assayed by the addition of 100 µl of 3,3',5,5'-tetramethylbenzidine substrate in the dark at room temperature for 30 min. The reaction was stopped by adding 50 µl per well of 0.1 M HCl. The absorbance at 450 nm was measured with a microplate reader. Data were analyzed using a standard curve with a log-logistic model.

Immunization of mice

K18-hACE2 transgenic mice were obtained from GemPharmatech Co. Ltd., Jiangsu Province. For antigen formulation, 25 µg of S_{wt}, S_{pan}, or S₁₅ proteins were mixed with an equal volume of Al(OH)₃ for each mouse. K18-hACE2 or BALB/c mice aged 6 to 7 weeks were immunized twice by intramuscular injection (100 µl in each left and right quadriceps femoris muscle per mouse) 2 weeks apart. For booster studies, animals were vaccinated three times, with 2 weeks separating each vaccination.

Histology analysis

Lung tissues from infected mice were isolated when animals met the prespecified euthanasia criteria or at day 14 after SARS-CoV-2 infection. Tissues were fixed and stained using a standard hematoxylin and eosin staining procedure. The slides were scanned by Wuhan Servicebio Technology Co. Ltd. The pathological score includes (i) hemorrhage, exudation, and mucous; (ii) alveolar septum thickening and consolidation; (iii) recruitment and infiltration of inflammatory immune cells; (iv) pulmonary emphysema; and (v) pulmonary edema. For each feature, the score related to the severity is as follows: 0 indicates that no pathological change was observed, 1 indicates mild pathological change, 2 indicates moderate pathological change, and 3 indicates severe pathological change. Scores of these five parameters were added to determine the cumulative pathological score of a lung lobe.

Structure determination

The SARS-CoV-2 S trimer profile (PDB: 6XR8) was downloaded from Protein Data Bank (PDB). SARS-CoV-2 variant mutations were mapped on the S trimer as shown in fig. S5B. The structural figures were obtained using the PyMOL Molecular Graphics System, version 1.0.

Evolutionary dynamics of SARS-CoV-2 variants

Evolutionary dynamics of SARS-CoV-2 variants from the ongoing COVID-19 pandemic were available at www.gisaid.org/phylogenetics/global/nextstrain/. Briefly, the visualization only contained 3815 genomes in a single view for performance and legibility reasons (56, 57). The frequencies of SARS-CoV-2 variants in

the first week of each month were recorded, and the analysis was performed using GraphPad 7.0.

Defining the emergence time of individual mutations

The peptide sequences of SARS-CoV-2 S protein were downloaded from GISAID until July 2022. The original data exactly matched the "FASTA header format" and were used for further analysis. MAFFT (v7.487) was used to align each peptide sequence to the reference sequence (Wuhan-Hu-1/2019), and the mutation information was parsed using a custom python script (DOI:10.5281/zenodo.7380031). We analyzed 46 kinds of monomer mutations contained in seven types of combined mutants and classified each sequence according to the submission date, which indicated the occurrence of these monomeric mutations (ignoring mutations in other locations). To improve the accuracy, we discarded a sequence if it was missing sequence information in any of annotated monomer mutation sites. For the statistics of the appearance time of the WT sequence, we required it to be entirely consistent with the reference sequence.

Statistical analysis

All raw, individual-level data are presented in data file S1. Mann-Whitney tests or two-way analyses of variance (ANOVAs) were used to test for statistical significance. Log-rank tests were used for survival curve analyses. Only *P* values of 0.05 or lower were considered statistically significant (*, $^{\#}P < 0.05$, **, $^{\#\#}P < 0.01$, and ***, $^{\#\#\#}P < 0.001$). GraphPad Prism 7 software (GraphPad Software) was used for statistical analysis.

Supplementary Materials

This PDF file includes:

Figs. S1 to S10

Tables S1 and S2

Other Supplementary Material for this

manuscript includes the following:

Data file S1

MDAR Reproducibility Checklist

[View/request a protocol for this paper from Bio-protocol.](#)

REFERENCES AND NOTES

- C. Huang, Y. Wang, X. Li, L. Ren, J. Zhao, Y. Hu, L. Zhang, G. Fan, J. Xu, X. Gu, Z. Cheng, T. Yu, J. Xia, Y. Wei, W. Wu, X. Xie, W. Yin, H. Li, M. Liu, Y. Xiao, H. Gao, L. Guo, J. Xie, G. Wang, R. Jiang, Z. Gao, Q. Jin, J. Wang, B. Cao, Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. *Lancet* **395**, 497–506 (2020).
- W. J. Guan, Z. Y. Ni, Y. Hu, W. H. Liang, C. Q. Ou, J. X. He, L. Liu, H. Shan, C. L. Lei, D. S. C. Hui, B. du, L. J. Li, G. Zeng, K. Y. Yuen, R. C. Chen, C. L. Tang, T. Wang, P. Y. Chen, J. Xiang, S. Y. Li, J. L. Wang, Z. J. Liang, Y. X. Peng, L. Wei, Y. Liu, Y. H. Hu, P. Peng, J. M. Wang, J. Y. Liu, Z. Chen, G. Li, Z. J. Zheng, S. Q. Qiu, J. Luo, C. J. Ye, S. Y. Zhu, N. S. Zhong; China Medical Treatment Expert Group for Covid-19, Clinical characteristics of coronavirus disease 2019 in China. *N. Engl. J. Med.* **382**, 1708–1720 (2020).
- L. Chen, W. Liu, Q. Zhang, K. Xu, G. Ye, W. Wu, Z. Sun, F. Liu, K. Wu, B. Zhong, Y. Mei, W. Zhang, Y. Chen, Y. Li, M. Shi, K. Lan, Y. Liu, RNA based mNGS approach identifies a novel human coronavirus from two individual pneumonia cases in 2019 Wuhan outbreak. *Emerg. Microbes Infect.* **9**, 313–319 (2020).
- World Health Organization, WHO Coronavirus (COVID-19) Dashboard; <https://covid19.who.int>.
- CNCB-NGDC, Visualization of Genomic Variations, Variation Annotation; <https://bigd.big.ac.cn/ncov/variation/annotation>.
- T. Kirby, New variant of SARS-CoV-2 in UK causes surge of COVID-19. *Lancet Respir. Med.* **9**, e20–e21 (2021).
- B. Zhou, T. T. N. Thao, D. Hoffmann, A. Taddeo, N. Ebert, F. Labrousseau, A. Pohlmann, J. King, S. Steiner, J. N. Kelly, J. Portmann, N. J. Halwe, L. Ulrich, B. S. Trüeb, X. Fan, B. Hoffmann, L. Wang, L. Thomann, X. Lin, H. Stalder, B. Pozzi, S. de Brot, N. Jiang, D. Cui, J. Hossain, M. M. Wilson, M. W. Keller, T. J. Stark, J. R. Barnes, R. Dijkman, J. Jores, C. Benarafa, D. E. Wentworth, V. Thiel, M. Beer, SARS-CoV-2 spike D614G change enhances replication and transmission. *Nature* **592**, 122–127 (2021).
- E. B. Hodcroft, M. Zuber, S. Nadeau, T. G. Vaughan, K. H. D. Crawford, C. L. Althaus, M. L. Reichmuth, J. E. Bowen, A. C. Walls, D. Corti, J. D. Bloom, D. Veessler, D. Mateo, A. Hernando, I. Comas, F. González-Candelas; SeqCOVID-SPAIN consortium, F. González-Candelas, G. A. Goig, Á. Chiner-Oms, I. Cancino-Muñoz, M. G. López, M. Torres-Puente, I. Gomez-Navarro, S. Jiménez-Serrano, L. Ruiz-Roldán, M. A. Bracho, N. García-González, L. Martínez-Priego, I. Galán-Vendrell, P. Ruiz-Hueso, G. de Marco, M. L. Ferrús, S. Carbó-Ramírez, G. D'Auria, M. Coscollá, P. Ruiz-Rodríguez, F. J. Roig-Sena, I. Sanmartín, D. García-Souto, A. Pequeno-Valtierra, J. M. C. Tubio, J. Rodríguez-Castro, N. Rabella, F. Navarro, E. Miró, M. Rodríguez-Iglesias, F. Galán-Sánchez, S. Rodríguez-Pallares, M. de Toro, M. B. Escudero, J. M. Azcona-Gutiérrez, M. B. Alberdi, A. Mayor, A. L. García-Basteiro, G. Moncunil, C. Dobaño, P. Cisteró, D. García-Viedma, L. Pérez-Lago, M. Herranz, J. Sicilia, P. Catalán-Alonso, P. Muñoz, C. Muñoz-Cuevas, G. Rodríguez-Rodríguez, J. Alberola-Enguadanos, J. M. Nogueira, J. J. Camarena, A. Rezusta, A. Tristanch-Baró, A. Milagro, N. F. Martínez-Cameo, Y. Gracia-Grataloup, E. Martró, A. E. Bordoy, A. Not, A. Antuori-Torres, R. Benito, S. Algarate, J. Bueno, J. L. del Pozo, J. A. Boga, C. Castelló-Abietar, S. Rojo-Alba, M. E. Alvarez-Argüelles, S. Melon, M. Aranzamendi-Zaldumbide, A. Vergara-Gómez, J. Fernández-Pinero, M. J. Martínez, J. Vila, E. Rubio, A. Peiró-Mestres, J. Navero-Castillejos, D. Posada, D. Valverde, N. Estévez-Gómez, I. Fernandez-Silva, L. de Chiara, P. Gallego-García, N. Varela, R. Moreno, M. D. Tirado, U. Gomez-Pinedo, M. Gozalo-Margüello, M. Eliecer-Cano, J. M. Méndez-Legaza, J. Rodríguez-Lozano, M. Siller, D. Pablo-Marcos, A. Oliver, J. Reina, C. López-Causapé, A. Canut-Blasco, S. Hernández-Crespo, M. L. A. Cordón, M. C. Lecároz-Agata, C. Gómez-González, A. Aguirre-Quinonero, J. I. López-Mirones, M. González-Torres, M. R. Almela-Ferrer, N. Gonzalo-Jiménez, M. M. Ruiz-García, A. Galiana, J. Sanchez-Almendo, G. Cilla, M. Montes, L. Piñero, A. Sorraín, J. M. Marimón, M. D. Gomez-Ruiz, J. L. López-Hontangas, E. M. González Barberá, J. M. Navarro-Marí, I. Pedrosa-Corral, S. Sanbonmatsu-Gámez, C. Pérez-González, F. Chamizo-López, A. Bordes-Benitez, D. Navarro, E. Albert, I. Torres, I. Gascón, C. J. Torregrosa-Hetland, E. Pastor-Bon, P. Cascales-Ramos, B. Fuster-Escrivá, C. Gimeno-Cardona, M. D. Ocete, R. Medina-Gonzalez, J. González-Cantó, O. Martínez-Macias, B. Palop-Borrás, I. de Toro, M. C. Mediavilla-Grado, M. Pérez-Ruiz, Ó. González-Recio, M. Gutiérrez-Rivas, E. Simarro-Córdoba, J. Lozano-Serra, L. Robles-Fonseca, A. de Salazar, L. Viñuela-González, N. Chueca, F. García, C. Gómez-Camarasa, A. Carvajal, R. de la Puente, V. Martín-Sánchez, J. M. Fregeneda-Grandes, A. J. Molina, H. Argüello, T. Fernández-Villa, M. A. Farga-Martí, V. Domínguez-Márquez, J. J. Costa-Alcalde, R. Trastoy, G. Barbeito-Casteñeiras, A. Coira, M. L. Pérez-del-Molino, A. Aguilera, A. M. Planas, A. Soriano, I. Fernandez-Cádenas, J. Pérez-Tur, M. Á. Marcos, A. Moreno-Docón, E. Viedma, J. Mingorance, J. C. Galán-Montemayor, M. Parra-Grande, T. Stadler, R. A. Neher, Spread of a SARS-CoV-2 variant through Europe in the summer of 2020. *Nature* **595**, 707–712 (2021).
- A. S. Lauring, E. B. Hodcroft, Genetic variants of SARS-CoV-2-what do they mean? *JAMA* **325**, 529–531 (2021).
- World Health Organization, Tracking SARS-CoV-2 variants; www.who.int/activities/tracking-SARS-CoV-2-variants/tracking-SARS-CoV-2-variants.
- cov-lineages.org, B.1.1.7; https://cov-lineages.org/global_report_B.1.1.7.html.
- Á. O'Toole, V. Hill, O. G. Pybus, A. Watts, I. I. Bogoch, K. Khan, J. P. Messina, The COVID-19 Genomics UK (COG-UK) consortium, Network for Genomic Surveillance in South Africa (NGS-SA), Brazil-UK CADDE Genomic Network, H. Tegally, R. R. Lessells, J. Gindhari, S. Pillay, K. A. Tumedi, G. Nyepetsi, M. Kebabonye, M. Matsheka, M. Mine, S. Tokajian, H. Hassan, T. Salloum, G. Merhi, J. Koweyes, J. L. Geoghegan, J. de Ligt, X. Ren, M. Storey, N. E. Freed, C. Pattabiraman, P. Prasad, A. S. Desai, R. Vasanthapuram, T. F. Schulz, L. Steinbrück, T. Stadler, Swiss Viollier Sequencing Consortium, A. Parisi, A. Bianco, D. García de Viedma, S. Buenestado-Serrano, V. Borges, J. Isidro, S. Duarte, J. P. Gomes, N. S. Zuckerman, M. Mandelboim, O. Mor, T. Seemann, A. Arnott, J. Draper, M. Gall, W. Rawlinson, I. Deveson, S. Schlebusch, J. McMahon, L. Leong, C. K. Lim, M. Chironna, D. Laconson, A. Bal, L. Josset, E. Holmes, K. St. George, E. Lasek-Nesselquist, R. S. Sikkema, B. B. Oude Munnink, M. Koopmans, M. Brytting, V. Sudha Rani, S. Pavan, T. Smura, A. Heim, S. Kurkela, M. Umair, M. Salman, B. Bartolini, M. Rueca, C. Drosten, T. Wolff, O. Silander, D. Eggink, C. Reusken, H. Vennema, A. Park, SEARCH Alliance San Diego, National Virus Reference Laboratory, SeqCOVID-Spain, Danish Covid-19 Genome Consortium (DCGC), Communicable Diseases Genomic Network (CDGN), Dutch National SARS-CoV-2 surveillance program, Division of Emerging Infectious Diseases KDCA, T. de Oliveira, N. R. Faria, A. Rambaut, M. U. G. Kraemer, Tracking the international spread of SARS-CoV-2 lineages B.1.1.7 and B.1.351/501Y-V2; <https://virological.org/t/tracking-the-international-spread-of-sars-cov-2-lineages-b-1-1-7-and-b-1-351-501y-v2/592>.

13. S. Pascarella, M. Ciccozzi, D. Zella, M. Bianchi, F. Benedetti, D. Benvenuto, F. Broccoli, R. Cauda, A. Caruso, S. Angeletti, M. Giovanetti, A. Cassone, SARS-CoV-2 B.1.617 Indian variants: Are electrostatic potential changes responsible for a higher transmission rate? *J. Med. Virol.* **93**, 6551–6556 (2021).
14. GISAID, Genomic epidemiology of SARS-CoV-2 with subsampling focused globally over the past 6 months; www.gisaid.org/phyloinformatics/global/nextstrain/.
15. M. Dai, M. Dai, W. Du, C. Martinez-Romero, T. Leenders, T. Wennekes, G. F. Rimmelzwaan, F. J. M. van Kuppeveld, R. A. M. Fouchier, A. Garcia-Sastre, E. de Vries, C. A. M. de Haana, Analysis of the evolution of pandemic influenza A(H1N1) virus neuraminidase reveals entanglement of different phenotypic characteristics. *MBio* **12**, e00287-21 (2021).
16. B. Korber, W. M. Fischer, S. Gnanakaran, H. Yoon, J. Theiler, W. Abfalterer, N. Hengartner, E. E. Giorgi, T. Bhattacharya, B. Foley, K. M. Hastie, M. D. Parker, D. G. Partridge, C. M. Evans, T. M. Freeman, T. I. de Silva, Tracking changes in SARS-CoV-2 spike: Evidence that D614G increases infectivity of the COVID-19 virus. *Cell* **182**, 812–827.e19 (2020).
17. J. Plante, Y. Liu, J. Liu, H. Xia, B. A. Johnson, K. G. Lokugamage, X. Zhang, A. E. Muruato, J. Zou, C. R. Fontes-Garfias, D. Mirchandani, D. Scharton, J. P. Bilello, Z. Ku, Z. An, B. Kalveram, A. N. Freiberg, V. D. Menachery, X. Xie, K. S. Plante, S. C. Weaver, P.-Y. Shi, Spike mutation D614G alters SARS-CoV-2 fitness. *Nature* **592**, 116–121 (2021).
18. K. Siddle, L. A. Krasinikova, G. K. Moreno, S. F. Schaffner, J. Vostok, N. A. Fitzgerald, J. E. Lemieux, N. Barkas, C. Loreth, I. Specht, C. H. Tomkins-Tinch, J. S. Paull, B. Schaeffer, B. P. Taylor, B. Loftness, H. Johnson, P. L. Schubert, H. M. Shephard, M. Doucette, T. Fink, A. S. Lang, S. Baez, J. Beauchamp, S. Hennigan, E. Buzby, S. Ash, J. Brown, S. Clancy, S. Cofsky, L. Gagne, J. Hall, R. Harrington, G. L. Gionet, K. C. De Ruff, M. E. Vozdak, G. C. Adams, S. T. Dobbins, S. D. Slack, S. K. Reilly, L. M. Anderson, M. C. Cipicchio, M. T. De Felice, J. L. Grimsby, S. E. Anderson, B. S. Blumenstiel, J. C. Meldrim, H. M. Rooke, G. Vicente, N. L. Smith, K. S. Messer, F. L. Reagan, Z. M. Mandese, M. D. Lee, M. C. Ray, M. E. Fisher, M. A. Ulcena, C. M. Nolet, S. E. English, K. L. Larkin, K. Vernest, S. Chaluvadi, D. Arvidson, M. Melchionno, T. Covell, V. Harik, T. Brock-Fisher, M. Dunn, A. Kearns, W. P. Hanage, C. Bernard, A. Philippakis, N. J. Lennon, S. B. Gabriel, G. R. Gallagher, S. Smole, L. C. Madoff, C. M. Brown, D. J. Park, B. L. Madnisi, P. C. Sabeti, Transmission from vaccinated individuals in a large SARS-CoV-2 Delta variant outbreak. *Cell* **185**, 485–492.e10 (2022).
19. H. Tseng, B. K. Ackerson, Y. Luo, L. S. Sy, C. A. Talarico, Y. Tian, K. J. Bruxvoort, J. E. Tubert, A. Florea, J. H. Ku, G. S. Lee, S. K. Choi, H. S. Takhar, M. Aragones, L. Qian, Effectiveness of mRNA-1273 against SARS-CoV-2 Omicron and Delta variants. *Nat. Med.* **28**, 1063–1071 (2022).
20. M. Kang, Y. Yi, Y. Li, L. Sun, A. Deng, T. Hu, J. Zhang, J. Liu, M. Cheng, S. Xie, M. Luo, J. Jiang, Y. Jiang, S. Tang, J. He, Effectiveness of inactivated COVID-19 vaccines against illness caused by the B.1.617.2 (Delta) variant during an outbreak in Guangdong, China: A cohort study. *Ann. Intern. Med.* **175**, 533–540 (2022).
21. V. Emami, R. Reddy, S. Goswami, Effectiveness of Covid-19 vaccines against the B.1.617.2 (Delta) variant. *N. Engl. J. Med.* **385**, e92 (2021).
22. R. Wang, Q. Zhang, J. Ge, W. Ren, R. Zhang, J. Lan, B. Ju, B. Su, F. Yu, P. Chen, H. Liao, Y. Feng, X. Li, X. Shi, Z. Zhang, F. Zhang, Q. Ding, T. Zhang, X. Wang, L. Zhang, Analysis of SARS-CoV-2 variant mutations reveals neutralization escape mechanisms and the ability to use ACE2 receptors from additional species. *Immunity* **54**, 1611–1621.e5 (2021).
23. I. A. T. M. Ferreira, S. A. Kemp, R. Datir, A. Saito, B. Meng, P. Rakshit, A. Takaori-Kondo, Y. Kosugi, K. Uriu, I. Kimura, K. Shirakawa, A. Abdullahi, A. Agarwal, S. Ozono, K. Tokunaga, K. Sato, R. K. Gupta; CITIID-NIHR BioResource COVID-19 Collaboration; Indian SARS-CoV-2 Genomics Consortium; Genotype to Phenotype Japan (G2P-Japan) Consortium, SARS-CoV-2 B.1.617 mutations L452 and E484Q are not synergistic for antibody evasion. *J. Infect. Dis.* **224**, 989–994 (2021).
24. W. F. Garcia-Beltran, E. C. Lam, K. S. Denis, A. D. Nitido, Z. H. Garcia, B. M. Hauser, J. Feldman, M. N. Pavlovic, D. J. Gregory, M. C. Poznansky, A. Sigal, A. G. Schmidt, A. J. Iafraite, V. Naranbhaj, A. B. Balazs, Multiple SARS-CoV-2 variants escape neutralization by vaccine-induced humoral immunity. *Cell* **184**, 2372–2383.e9 (2021).
25. M. Massard, R. Eftimie, A. Perasso, B. Sausseureau, A multi-strain epidemic model for COVID-19 with infected and asymptomatic cases: Application to French data. *J. Theor. Biol.* **545**, 111117 (2022).
26. Y. Liu, J. Rocklöv, The effective reproductive number of the Omicron variant of SARS-CoV-2 is several times relative to Delta. *J. Travel Med.* **29**, (2022).
27. A. Wilhelm, M. Wiedera, K. Grikscheit, T. Toptan, B. Schenk, C. Pallas, M. Metzler, N. Kohmer, S. Hoehl, R. Marschalek, E. Herrmann, F. A. Helfritz, T. Wolf, U. Goetsch, S. Ciesek, Limited neutralisation of the SARS-CoV-2 Omicron subvariants BA.1 and BA.2 by convalescent and vaccine serum and monoclonal antibodies. *EBioMedicine* **82**, 104158 (2022).
28. S. Cele, L. Jackson, D. S. Khoury, K. Khan, T. Moyo-Gwete, H. Tegally, J. E. San, D. Cromer, C. Scheepers, D. G. Amoako, F. Karim, M. Bernstein, G. Lustig, D. Archary, M. Smith, Y. Ganga, Z. Jule, K. Reedy, S.-H. Hwa, J. Giandhari, J. M. Blackburn, B. I. Gosnell, S. S. Abdool Karim, W. Hanekom; NGS-SA; COMMIT-KZN Team, A. von Gottberg, J. N. Bhiman, R. J. Lessells, M.-Y. S. Moosa, M. P. Davenport, T. de Oliveira, P. L. Moore, A. Sigal, Omicron extensively but incompletely escapes Pfizer BNT162b2 neutralization. *Nature* **602**, 654–656 (2022).
29. X. Zhao, D. Li, W. Ruan, Z. Chen, R. Zhang, A. Zheng, S. Qiao, X. Zheng, Y. Zhao, L. Dai, P. Han, G. F. Gao, Effects of a prolonged booster interval on neutralization of omicron variant. *N. Engl. J. Med.* **386**, 894–896 (2022).
30. Z. Liu, J. Zhou, W. Xu, W. Deng, Y. Wang, M. Wang, Q. Wang, M. Hsieh, J. Dong, X. Wang, W. Huang, L. Xing, M. He, C. Tao, Y. Xie, Y. Zhang, Y. Wang, J. Zhao, Z. Yuan, C. Qin, S. Jiang, L. Lu, A novel STING agonist-adjuvanted pan-sarbecovirus vaccine elicits potent and durable neutralizing antibody and T cell responses in mice, rabbits and NHPs. *Cell Res.* **32**, 269–287 (2022).
31. Q. Liu, Q. Xiong, F. Mei, C. Ma, Z. Zhang, B. Hu, J. Xu, Y. Jiang, F. Zhan, S. Zhou, L. Tao, X. Chen, M. Guo, X. Wang, Y. Fang, S. Shen, Y. Liu, F. Liu, L. Zhou, K. Xu, C. Ke, F. Deng, K. Cai, H. Yan, Y. Chen, K. Lan, Antibody neutralization to SARS-CoV-2 and variants after 1 year in Wuhan, China. *Innovation* **3**, 100181 (2022).
32. Q. Li, J. Nie, J. Wu, L. Zhang, R. Ding, H. Wang, Y. Zhang, T. Li, S. Liu, M. Zhang, C. Zhao, H. Liu, L. Nie, H. Qin, M. Wang, Q. Lu, X. Li, J. Liu, H. Liang, Y. Shi, Y. Shen, L. Xie, L. Zhang, X. Qu, W. Xu, W. Huang, Y. Wang, SARS-CoV-2 501Y.V2 variants lack higher infectivity but do have immune escape. *Cell* **184**, 2362–2371.e9 (2021).
33. M. Hoffmann, K. Mösbauer, H. Hofmann-Winkler, A. Kaul, H. Kleine-Weber, N. Krüger, N. C. Gassen, M. A. Müller, C. Drosten, S. Pöhlmann, Chloroquine does not inhibit infection of human lung cells with SARS-CoV-2. *Nature* **440**, 274 (2006).
34. Z. Gu, R. Eils, M. Schlesner, Complex heatmaps reveal patterns and correlations in multi-dimensional genomic data. *Bioinformatics* **32**, 2847–2849 (2016).
35. K. Corbett, M. Gagne, D. A. Wagner, S. O'Connell, S. R. Narpala, D. R. Flebbe, S. F. Andrew, R. L. Davis, B. Flynn, T. S. Johnston, C. D. Stringham, L. Lai, D. Valentin, A. Van Ry, Z. Flinchbaugh, A. P. Werner, J. I. Molina, M. Sripama, S. O'Dell, S. D. Schmidt, C. Tucker, A. Choi, M. Koch, K. W. Bock, M. Minai, B. M. Nagata, G. S. Alvarado, A. R. Henry, F. Laboune, C. A. Schramm, Y. Zhang, E. S. Yang, L. Wang, M. Choe, S. Boyoglu-Barnum, S. Wei, E. Lamb, S. T. Nurmukhambetova, S. J. Provost, M. M. Donaldson, J. Marquez, J.-P. M. Todd, A. Cook, A. Dodson, A. Pekosz, E. Boritz, A. Ploquin, N. Doria-Rose, L. Pessaint, H. Andersen, K. E. Foulds, J. Misasi, K. Wu, A. Carfi, M. C. Nason, J. Mascola, I. N. Moore, D. K. Edwards, M. G. Lewis, M. S. Suthar, M. Roederer, A. M. Dermott, D. C. Douek, N. J. Sullivan, B. S. Graham, R. A. Seder, Protection against SARS-CoV-2 beta variant in mRNA-1273 vaccine-boostered nonhuman primates. *Science*, eabl8912 (2021).
36. L. Bai, Y. Zhao, J. Dong, S. Liang, M. Guo, X. Liu, X. Wang, Z. Huang, X. Sun, Z. Zhang, L. Dong, Q. Liu, Y. Zheng, D. Niu, M. Xiang, K. Song, J. Ye, W. Zheng, Z. Tang, M. Tang, Y. Zhou, C. Shen, M. Dai, L. Zhou, Y. Chen, H. Yan, K. Lan, K. Xu, Coinfection with influenza A virus enhances SARS-CoV-2 infectivity. *Cell Res.* **31**, 395–403 (2021).
37. Y. Weisblum, F. Schmidt, F. Zhang, J. DaSilva, D. Poston, J. C. C. Lorenzi, F. Muecksch, M. Rutkowska, H. H. Hoffmann, E. Michailidis, C. Gaebler, M. Agudelo, A. Cho, Z. Wang, A. Gazumyan, M. Cipolla, L. Luchsinger, C. D. Hillyer, M. Caskey, D. F. Robbiani, C. M. Rice, M. C. Nussenzweig, T. Hatziioannou, P. D. Bieniasz, Escape from neutralizing antibodies by SARS-CoV-2 spike protein variants. *eLife* **9**, e61312 (2020).
38. E. Andreano, G. Piccini, D. Licastro, L. Casalino, N. V. Johnson, I. Paciello, S. Dal Monego, E. Pantano, N. Manganaro, A. Manenti, R. Manna, E. Casa, I. Hyseni, L. Benincasa, E. Montomoli, R. E. Amaro, J. S. McLellan, R. Rappuoli, SARS-CoV-2 escape in vitro from a highly neutralizing COVID-19 convalescent plasma. *bioRxiv* 2020.12.28.424451 [Preprint]. 28 December 2020. <https://doi.org/10.1101/2020.12.28.424451>.
39. C. Motozono, M. Toyoda, J. Zahradnik, A. Saito, H. Nasser, T. S. Tan, I. Ngare, I. Kimura, K. Uriu, Y. Kosugi, Y. Yue, R. Shimizu, J. Ito, S. Torii, A. Yonekawa, N. Shimono, Y. Nagasaki, R. Minami, T. Toya, N. Sekiya, T. Fukuhara, Y. Matsuura, G. Schreiber; Genotype to Phenotype Japan (GP-Japan) Consortium, T. Ikeda, S. Nakagawa, T. Ueno, K. Sato, SARS-CoV-2 spike L452R variant evades cellular immunity and increases infectivity. *Cell Host Microbe* **29**, 1124–1136.e11 (2021).
40. A. Greaney, A. N. Loes, K. H. D. Crawford, T. N. Starr, K. D. Malone, H. Y. Chu, J. D. Bloom, Comprehensive mapping of mutations in the SARS-CoV-2 receptor-binding domain that affect recognition by polyclonal human plasma antibodies. *Cell Host Microbe* **29**, 463–476.e6 (2021).
41. Q. Wang, Y. Guo, S. Ikhetani, M. S. Nair, Z. Li, H. Mohri, M. Wang, J. Yu, A. D. Bowen, J. Y. Chang, J. G. Shah, N. Nguyen, Z. Chen, K. Meyers, M. T. Yin, M. E. Sobieszczyk, Z. Sheng, Y. Huang, L. Liu, D. D. Ho, Antibody evasion by SARS-CoV-2 Omicron subvariants BA.2.12.1, BA.4, and BA.5. *Nature* **608**, 603–608 (2022).
42. S. Liu, G. Xiao, Y. Chen, Y. He, J. Niu, C. R. Escalante, H. Xiong, J. Farmer, A. K. Debnath, P. Tien, S. Jiang, Interaction between heptad repeat 1 and 2 regions in spike protein of SARS-associated coronavirus: Implications for virus fusogenic mechanism and identification of fusion inhibitors. *Lancet* **363**, 938–947 (2004).
43. S. Xia, M. Liu, C. Wang, W. Xu, Q. Lan, S. Feng, F. Qi, L. Bao, L. du, S. Liu, C. Qin, F. Sun, Z. Shi, Y. Zhu, S. Jiang, L. Lu, Inhibition of SARS-CoV-2 (previously 2019-nCoV) infection by a highly potent pan-coronavirus fusion inhibitor targeting its spike protein that harbors a high capacity to mediate membrane fusion. *Cell Res.* **30**, 343–355 (2020).
44. H. Zhao, K. K. W. To, H. Lam, X. Zhou, J. F. W. Chan, Z. Peng, A. C. Y. Lee, J. Cai, W. M. Chan, J. D. Ip, C. C. S. Chan, M. L. Yeung, A. J. Zhang, A. W. H. Chu, S. Jiang, K. Y. Yuen, Cross-linking

- peptide and repurposed drugs inhibit both entry pathways of SARS-CoV-2. *Nat. Commun.* **12**, 1517 (2021).
45. L. Sun, B. W. Y. Mok, L. L. Chen, J. M. C. Chan, O. T. Y. Tsang, B. H. S. Lam, V. W. M. Chuang, A. W. H. Chu, W. M. Chan, J. D. Ip, B. P. C. Chan, R. Zhang, C. C. Y. Yip, V. C. C. Cheng, K. H. Chan, D. Y. Jin, I. F. N. Hung, K. Y. Yuen, H. Chen, K. Kai Wang To, Neutralization of SARS-CoV-2 Omicron variant by sera from BNT162b2 or Coronavac vaccine recipients. *Clin. Infect. Dis.* **75**, e822–e826 (2021).
 46. J. Ai, H. Zhang, Q. Zhang, Y. Zhang, K. Lin, Z. Fu, J. Song, Y. Zhao, M. Fan, H. Wang, C. Qiu, Y. Zhou, W. Zhang, Recombinant protein subunit vaccine booster following two-dose inactivated vaccines dramatically enhanced anti-RBD responses and neutralizing titers against SARS-CoV-2 and Variants of Concern. *Cell Res.* **32**, 103–106 (2022).
 47. X. Liu, Y. Li, Z. Wang, S. Cao, W. Huang, L. Yuan, Y.-J. Huang, Y. Zheng, J. Chen, B. Ying, Z. Xiang, J. Shi, J. Zhao, Z. Huang, C.-F. Qin, Safety and superior immunogenicity of heterologous boosting with an RBD-based SARS-CoV-2 mRNA vaccine in Chinese adults. *Cell Res.* **32**, 777–780 (2022).
 48. Y. Cao, X. Hao, X. Wang, Q. Wu, R. Song, D. Zhao, W. Song, Y. Wang, A. Yisimayi, W. Wang, W. Zhang, J. du, H. Yu, X. S. Xie, R. Jin, Humoral immunogenicity and reactogenicity of CoronaVac or ZF2001 booster after two doses of inactivated vaccine. *Cell Res.* **32**, 107–109 (2022).
 49. X. Zhao, R. Zhang, S. Qiao, X. Wang, W. Zhang, W. Ruan, L. Dai, P. Han, G. F. Gao, Omicron SARS-CoV-2 neutralization from inactivated and ZF2001 vaccines. *N. Engl. J. Med.* **387**, 277–280 (2022).
 50. L. Qu, Z. Yi, Y. Shen, L. Lin, F. Chen, Y. Xu, Z. Wu, H. Tang, X. Zhang, F. Tian, C. Wang, X. Xiao, X. Dong, L. Guo, S. Lu, C. Yang, C. Tang, Y. Yang, W. Yu, J. Wang, Y. Zhou, Q. Huang, A. Yisimayi, S. Liu, W. Huang, Y. Cao, Y. Wang, Z. Zhou, X. Peng, J. Wang, X. S. Xie, W. Wei, Circular RNA vaccines against SARS-CoV-2 and emerging variants. *Cell* **185**, 1728–1744.e16 (2022).
 51. A. Cohen, P. N. P. Gnanapragasam, Y. E. Lee, P. R. Hoffman, S. Ou, L. M. Kakutani, J. R. Keefe, H.-J. Wu, M. Howarth, A. P. West, C. O. Barnes, M. C. Nussenzweig, P. J. Bjorkman, Mosaic nanoparticles elicit cross-reactive immune responses to zoonotic coronaviruses in mice. *Science* **371**, 735–741 (2021).
 52. Q. Li, M. Zhang, Z. Liang, L. Zhang, X. Wu, C. Yang, Y. An, J. Tong, S. Liu, T. Li, Q. Cui, J. Nie, J. Wu, W. Huang, Y. Wang, Antigenicity comparison of SARS-CoV-2 Omicron sublineages with other variants contained multiple mutations in RBD. *MedComm* **3**, e130 (2022).
 53. H.-L. Xiong, Y.-T. Wu, J.-L. Cao, R. Yang, Y.-X. Liu, J. Ma, X.-Y. Qiao, X.-Y. Yao, B.-H. Zhang, Y.-L. Zhang, W.-H. Hou, Y. Shi, J.-J. Xu, L. Zhang, S.-J. Wang, B.-R. Fu, T. Yang, S.-X. Ge, J. Zhang, Q. Yuan, B.-Y. Huang, Z.-Y. Li, T.-Y. Zhang, N.-S. Xia, Robust neutralization assay based on SARS-CoV-2 S-protein-bearing vesicular stomatitis virus (VSV) pseudovirus and ACE2-overexpressing BHK21 cells. *Emerg. Microbes Infect.* **9**, 2105–2113 (2020).
 54. J. Nie, Q. Li, J. Wu, C. Zhao, H. Hao, H. Liu, L. Zhang, L. Nie, H. Qin, M. Wang, Q. Lu, X. Li, Q. Sun, J. Liu, C. Fan, W. Huang, M. Xu, Y. Wang, Quantification of SARS-CoV-2 neutralizing antibody by a pseudotyped virus-based assay. *Nat. Protoc.* **15**, 3699–3715 (2020).
 55. J. Hadfield, C. Megill, S. M. Bell, J. Huddleston, B. Potter, C. Callender, P. Sagulenko, T. Bedford, R. A. Neher, Nextstrain: Real-time tracking of pathogen evolution. *Bioinformatics* **34**, 4121–4123 (2018).
 56. P. Sagulenko, V. Puller, R. A. Neher, TreeTime: Maximum-likelihood phylodynamic analysis. *Virus Evol.* **4**, vex042 (2018).

Acknowledgments: We thank the members of the SARS-CoV-2 Vaccine Task Force Group for their efforts to support the vaccine development: D. Tang, L. Qu, Y. Xu, Q. Liu, X. Chen, M. Guo, X. Wang, Z. Huang, M. Dai, X. Wu, Q. Xiong, Y. Zhong, R. Sun, Y. Li, F. Liu, C. Zhou, and Y. Zhang. **Funding:** This work was supported in part by the National Natural Science Foundation of China (grant 32188101 to K.L. and grants 31922004 and 82272307 to K.X.), the Innovation Team Research Program of Hubei Province (2020CFA015 to K.L. and K.X.), and the Application and Frontier Research Program of the Wuhan Government (2019020701011463 to K.X.). We are grateful to a special fund for COVID-19 Research of Wuhan University, funds from the Beijing Taikang Yicai Foundation, and the Fundamental Research Funds for the Central Universities (2042022kf1188) for their support of this work. **Author contributions:** K.L. and K.X. conceived the project and designed the experiments. Y. Zhao, L.D., W.N., S.L., D.Z., D.N., M.X., and Q.Z. constructed the mutation plasmids of SARS-CoV-2 S protein and produced the pseudoviruses. L.D., D.N., and Q.Z. titrated and evaluated the infectivity of pseudoviruses. K.X., S.L., and Z.C. designed the pan-vaccine, and S.L. calculated the consensus sequence. W.N. and Z.C. performed vaccine immunization of mice and collected the serum samples. Y. Zhao, W.N., M.X., Z.C., and W.G. elucidated the immune escape of pseudoviruses against monoclonal antibodies and vaccine serum samples. Y. Zhou and D.W. performed the bioinformatics analysis of variants. H.Y., Y.C., Y.P., X.W., Y.Y., Z.J., and Y.J. provided technical support and materials. Y. Zhao, L.D., Z.C., K.X., and K.L. wrote the manuscript with input from other authors. Y. Zheng, S.L., and Z.Z. performed the live virus experiments in the ABSL-3 laboratory. All authors read and approved the final manuscript. **Competing interests:** Y.P., X.W., Y.Y., and Z.J. are employed by Wuhan Institute of Biological Products Co. Ltd. K.X. and K.L. are co-inventors on a patent titled “Novel coronavirus mutant strain S protein and novel coronavirus mutant strain subunit vaccine” (ZL202111181856.X and PCT/CN2022/086181). **Data and materials availability:** All data associated with this study are present in the paper or the Supplementary Materials. This work is licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. To view a copy of this license, visit <http://creativecommons.org/licenses/by/4.0/>. This license does not apply to figures/photos/artwork or other content included in the article that is credited to a third party; obtain authorization from the rights holder before using this material.

Submitted 27 January 2022
Resubmitted 23 September 2022
Accepted 30 November 2022
Published 4 January 2023
10.1126/scitranslmed.abo3332

Vaccination with S_{pan}, an antigen guided by SARS-CoV-2 S protein evolution, protects against challenge with viral variants in mice

Yongliang Zhao, Wenjia Ni, Simeng Liang, Lianghui Dong, Min Xiang, Zeng Cai, Danping Niu, Qiuhan Zhang, Dehe Wang, Yucheng Zheng, Zhen Zhang, Dan Zhou, Wenhua Guo, Yongbing Pan, Xiaoli Wu, Yimin Yang, Zhaoifei Jing, Yongzhong Jiang, Yu Chen, Huan Yan, Yu Zhou, Ke Xu, and Ke Lan

Sci. Transl. Med. **15** (677), eabo3332. DOI: 10.1126/scitranslmed.abo3332

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